

Topics: 14.6 Directional Derivatives and the Gradient Vector; 14.7 Maximum and Minimum Values
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- Find the directional derivative of the function at the given point in the indicated direction.
 - $f(x, y) = \frac{x}{y^2}$ at the point $P(3, -1)$, in the direction from P toward $Q(-2, 11)$.
 - $g(x, y) = x \ln y$ at the point $(2, e)$, in the direction that makes an angle of $\frac{\pi}{6}$ above the positive x -axis.
 - $h(x, y, z) = x^2y^3 - 4xz$ at the point $(1, 1, 1)$ in the direction of the vector $\mathbf{v} = \langle -1, 2, 0 \rangle$.
- Find the maximum rate of change of the function at the given point and the direction in which it occurs.
 - $f(x, y) = \sin(xy)$ at the point $(1, 0)$.
 - $g(x, y, z) = e^{2x} \cos(y - 2z)$ at the point $(1, \pi, 0)$.
- Find an equation of the tangent plane and parametric equations of the normal line to the surface $xy^2z^3 = 8$ at the point $P(2, 2, 1)$.
- For each function, find all critical points. Then use the Second Derivatives Test to determine whether the function has a local maximum, a local minimum, or a saddle point there.
 - $f(x, y) = 4x^2 + 9y^2 + 8x - 36y + 24$
 - $g(x, y) = (y - 2)x^2 - y^2$
 - $h(x, y) = (x + y^2)e^x$
- Find the absolute maximum and absolute minimum values of the function f on the set D .
 - $f(x, y) = x^2 + y^2 + x^2y + 9$, where $D = \{(x, y) \in \mathbb{R}^2 : |x| \leq 1 \text{ and } |y| \leq 1\}$
 - $f(x, y) = x^2 + y^2 - 2x$, where D is the closed triangular region with vertices $(2, 0)$, $(0, 2)$, and $(0, -2)$.

- A tennis ball manufacturer's monthly profit (in thousands of dollars) is modeled by

$$f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2,$$

where x is the number of tennis balls manufactured (in thousands) and y is the number of hours of advertising purchased. The company can produce at most 50,000 tennis balls per month, and it can purchase at most 25 hours of advertising. Find the values of x and y that maximize profit, and find the maximum profit.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Directional Derivatives. Let f be differentiable at a point P , and let $\mathbf{u}$ be a unit vector. The directional derivative of f at P in the direction of $\mathbf{u}$ is given by

$$D_{\mathbf{u}}f(P) = \nabla f(P) \cdot \mathbf{u}.$$

The directional derivative is the rate of change of f at P in the direction of $\mathbf{u}$. If $\nabla f(P) \neq \mathbf{0}$, then the maximum value of the directional derivative at P is $|\nabla f(P)|$ and this maximum occurs when $\mathbf{u}$ has the same direction as $\nabla f(P)$. In other words, $\nabla f(P)$ points in the direction of the fastest increase of f at P .

Tangent Planes to Level Surfaces. Suppose a surface S is expressed as a level surface

$$F(x, y, z) = k,$$

where F is differentiable, and let (x_0, y_0, z_0) be a point on S . Then the gradient vector $\nabla F(x_0, y_0, z_0)$ is orthogonal to the level surface at (x_0, y_0, z_0) , provided that $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$. Therefore, the tangent plane to the surface at (x_0, y_0, z_0) is the plane through that point with normal vector $\nabla F(x_0, y_0, z_0)$, and its equation is

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0.$$

Critical Points and Local Extrema. Let f be a function of two variables, and let (a, b) be a point in the domain of f . We say that f has a *local maximum* at (a, b) if $f(a, b) \geq f(x, y)$ for all points (x, y) near (a, b) . A *local minimum* is defined analogously. We say that (a, b) is a *critical point* of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if at least one of these partial derivatives does not exist at (a, b) . If f has a local maximum or local minimum at (a, b) , then (a, b) is necessarily a critical point. A point (a, b) at which $f_x(a, b) = f_y(a, b) = 0$ and f has neither a local maximum nor a local minimum is called a *saddle point*.

The Second Derivatives Test. Suppose that $f_x(a, b) = f_y(a, b) = 0$ and that the second partial derivatives of f are continuous near (a, b) . Define

$$D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2.$$

Then:

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .
- If $D(a, b) < 0$, then f has a saddle point at (a, b) .
- If $D(a, b) = 0$, the test is inconclusive.

Absolute Extrema on Closed and Bounded Sets. If f is continuous on a closed and bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value and an absolute minimum value on D . To find these absolute extrema, evaluate f at all critical points in the interior of D , and at all candidates for extrema on the boundary of D . The largest of these values is the absolute maximum value of f on D , and the smallest is the absolute minimum value.

Suggested Textbook Problems

Section 14.6: 1-44, 47-56

Section 14.7: 1-56